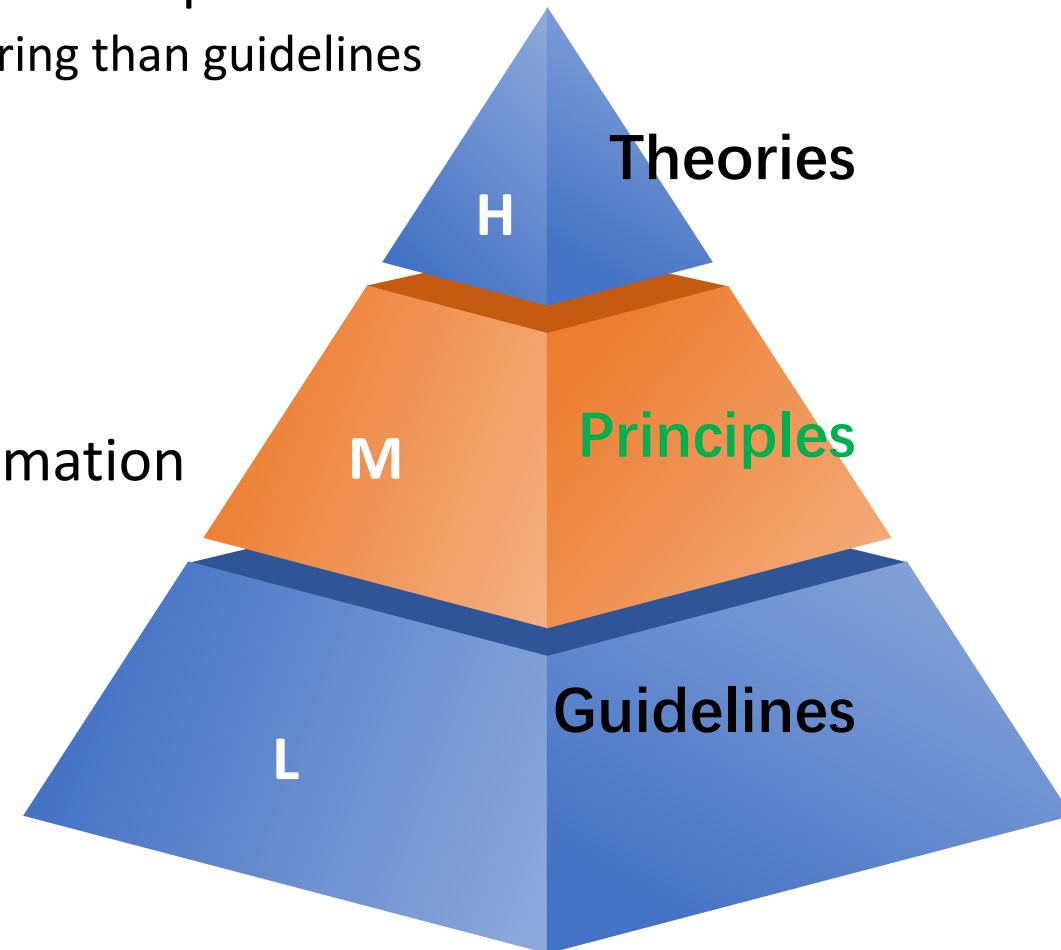


Principles

- Basics
 - Middle-level strategies or rules to analyse and compare UID alternatives
 - More fundamental, widely applicable, and enduring than guidelines
 - Determine users' skill levels
 - Identify the tasks
 - Choose an interaction style
 - Eight golden rules of UID
 - Ensure human control while increasing automation



Principles

- Determine Users' Skill Levels
 - **Learning about the users** is a simple idea but a difficult and unfortunately often underevaluated goal
 - All UID should begin with an **understanding of the users**, including population profiles that reflect their age, gender, physical and cognitive abilities, education, cultural backgrounds, training motivation, goals, and personality
 - Successful UIDers are aware that users learn, think, and solve problems in different ways
 - There are often several communities of users (e.g. various user personas, user groups from different countries) for an UI
 - Recall individual diversity and group/community diversity
 - The process of **getting to know the users** is never-ending, because there is so much to know and because the users keep changing

Sensitive observation

- Openness, curiosity, ...



Dialectic (positive<->negative) thinking

- Inverse thinking, perspective taking, essence-grasping thinking, ...

Principles

- Determine Users' Skill Levels
 - Besides general understanding of users, an **understanding of users' skills** with UIs and with the application domain is important
 - Users might be tested for their familiarity with UI features (E.g. traversing hierarchical menus or drawing tools)
 - Other tests might cover domain-specific abilities (E.g. knowledge of airport city codes, traffic management terminology)
 - Users' skill levels are directly related to quality of UI usage/interaction
 - A generic separation/categorization in terms of users' skills
 - Novice or first-time users
 - Knowledgeable intermittent users
 - Expert frequent users

CITY	3-LETTER CITY CODE	COUNTRY	AIRPORT CODE
Shanghai	SHA	China	PVG
Shenzhen	SZX	China	

Principles

- Determine Users' Skill Levels

- Novice or first-time users

- True novice users are assumed to know little of the task or UI concepts
 - First-time users are often professionals who know the task concepts well but have shallow knowledge of the UI concepts (E.g. a business traveler using a new rental car's navigation system)

- Guidelines
- Overcome their uncertainties via instructions or dialog boxes
 - Restrict vocabulary to a small number of familiar, consistently used concept terms
 - Informative feedback about accomplishment of each task is helpful and constructive
 - Specific error messages about mistakes by users is necessary
 - Carefully designed video demonstrations and online tutorials may be effective

- Knowledgeable intermittent users

- Expert frequent users

Principles

- Determine Users' Skill Levels

- Novice or first-time users
- Knowledgeable intermittent users
 - They have stable task concepts and broad knowledge of UI concepts, but may have difficulty in retaining the structure of menus or the location of features (E.g. business travelers filling for travel reimbursements, university administrative procedures)
- Expert frequent users

Guidelines

- The burden on their memories will be lightened by orderly structure in the menus
- Their retention over time can be facilitated by consistent terminology
- Realize UIs that emphasizes **recognition** rather than recall
- Protection from danger is necessary to support relaxed exploration of features and usage of partially forgotten action sequences
- They will benefit from context-dependent help to fill in missing pieces of task or UI knowledge

Principles

• Determine Users’ Skill Levels

- Novice or first-time users
- Knowledgeable intermittent users
- Expert frequent users
 - Expert “power” users are thoroughly familiar with the task and UI concepts and seek to get their work done quickly
 - They welcome shortcuts to carry out actions with just a few clicks or gestures
 - They demand rapid response times
 - They prefer brief and non-distracting feedback

```
9812 Given two generic  $m$ -by- $n$  time-variant matrices
9813
9814 \begin{align*}
9815 \mathbf{L} \equiv \begin{bmatrix} l_{ij}(t) \end{bmatrix} \quad \mathbf{H} \equiv \begin{bmatrix} h_{ij}(t) \end{bmatrix}
9816 \end{align*}
9817 that satisfies
9818 \begin{align*}
9819 \mathbf{L} \preceq \mathbf{H} \iff \mathbf{H} \succeq \mathbf{L},
9820 \end{align*}
9821 define the interval matrix set or for short the interval matrix
9822 \begin{equation} \text{\label{eq:interval_matrix}}
9823 \{\mathbf{L}, \mathbf{H}\} \equiv \{ \mathbf{X} \mid \mathbf{L} \preceq \mathbf{X} \preceq \mathbf{H} \}.
9824 \end{equation}
9825
9826 In practical applications, control system uncertainty may be unknown, yet it is usually somehow bounded. For example, given the generic
9827 formalism of linear state-space modelling with uncertainty described in (\ref{eq:uncertain_state_space_linear2})
9828 \begin{align*}
9829 \frac{d}{dt} \mathbf{x}(t) = (\mathbf{A} + \Delta \mathbf{A}) \mathbf{x}(t) + (\mathbf{B} + \Delta \mathbf{B}) \mathbf{u}(t),
9830 \end{align*}
9831 the uncertainty factors  $\Delta \mathbf{A}$  and  $\Delta \mathbf{B}$  are at least bounded between their respective element-wise minimum
9832 value matrix and element-wise maximum value matrix, namely
9833 \begin{subequations} \text{\label{eq:A+DA_B+DB_interval_matrix1}}
9834 \begin{align}
9835 \underline{\Delta \mathbf{A}} \preceq \Delta \mathbf{A} \preceq \overline{\Delta \mathbf{A}}, \quad \underline{\Delta \mathbf{B}} \preceq \Delta \mathbf{B} \preceq \overline{\Delta \mathbf{B}}
9836 \end{align}
9837 or equivalently
9838 \begin{subequations} \text{\label{eq:A+DA_B+DB_interval_matrix2}}
9839 \begin{align}
9840 \mathbf{A} + \underline{\Delta \mathbf{A}} \preceq \mathbf{A} + \Delta \mathbf{A} \preceq \mathbf{A} + \overline{\Delta \mathbf{A}}, \quad \mathbf{B} + \underline{\Delta \mathbf{B}} \preceq \mathbf{B} + \Delta \mathbf{B} \preceq \mathbf{B} + \overline{\Delta \mathbf{B}}.
9841 \end{align}
9842 \end{subequations}
9843
```

ESC: 取消操作。	Ctrl+I: 斜体。	Ctrl+F10: 将文档窗口最大化。
Home: 定位到该行最左侧。	Ctrl+N: 新建文档。	Ctrl+F12: 快速「打开」。
End: 定位到该行最右侧。	Ctrl+U: 为字符添加下划线。	Ctrl+]: 增大字号。
Shift: 连续/快速选择。	Ctrl+A: 全选内容。	Ctrl+[: 减小字号。
Tab: 缩进对齐。	Ctrl+X: 剪切所选文本或对象。	Ctrl+= (等号): 应用下标格式。
Alt+: 竖向选择文本。	Ctrl+V: 粘贴所选文本或对象。	Ctrl+Home: 快速回到文档头。
鼠标单击+: 选中一行。	Ctrl+Z: 撤销上一操作。	Ctrl+Enter: 另起一页。
鼠标双击+: 选中一段。	Ctrl+Y: 后退一步。	Shift+F3: 改变字母大小写。
鼠标三击+: 选中全文。	Ctrl+O: 打开最近文档。	Shift+F4: 重复「查找」或「定位」操作。
F1: 调出帮助窗口。	Ctrl+D: 快速调用字体对话框。	Alt+F7: 快速定位语法错误。
F4: 重复上一步操作。	Ctrl+E: 段落居中对齐。	Alt+F8: 调用宏窗口。
F5: 调出定位窗口。	Ctrl+J: 两端对齐。	Alt+F11: 显示·Microsoft·Visual·Basic·代码。
F7: 快速检查语法。	Ctrl+L: 左对齐。	Alt+=: 快速调用公式。
F9: 更新选定的域。	Ctrl+R: 右对齐。	Alt+P+: 不打开文件就能看到里面的内容。
F12: 文件另存为。	Ctrl+F: 快速查找。	Ctrl+Shift+C/V: 复制/粘贴格式。
Ctrl+1: 单倍行距。	Ctrl+H: 替换内容。	Ctrl+Shift+G: 查看字数统计。
Ctrl+2: 两倍行距。	Ctrl+P: 快速打开打印窗口。	Ctrl+Shift++ (加号): 应用上标格式。
Ctrl+5: 1.5·倍行距。	Ctrl+Q: 取消段落格式。	Ctrl+Shift+Z: 取消人工设置的文本格式。
Ctrl+0: 在段前添加一行间距。	Ctrl+S: 保存。	Ctrl+Shift+8: 显示/隐藏标记符号。
Ctrl+B: 加粗。	Ctrl+W: 关闭当前文档。	

Given two generic m -by- n time-variant matrices

$$\mathbf{L} \equiv [l_{ij}(t)], \quad \mathbf{H} \equiv [h_{ij}(t)]$$

that satisfies

$$\mathbf{L} \preceq \mathbf{H} \iff \mathbf{H} \succeq \mathbf{L},$$

define the **interval matrix set** or for short the **interval matrix**

$$[\mathbf{L}, \mathbf{H}] \equiv \{\mathbf{X} \mid \mathbf{L} \preceq \mathbf{X} \preceq \mathbf{H}\}. \tag{5.24}$$

In practical applications, control system uncertainty may be unknown, yet it is usually somehow bounded. For example, given the generic formalism of linear state-space modelling with uncertainty described in (5.14)

$$\frac{d}{dt} \mathbf{x} = (\mathbf{A} + \Delta \mathbf{A}) \mathbf{x} + (\mathbf{B} + \Delta \mathbf{B}) \mathbf{u},$$

the uncertainty factors $\Delta \mathbf{A}$ and $\Delta \mathbf{B}$ are at least bounded between their respective element-wise minimum value matrix and element-wise maximum value matrix, namely

$$\underline{\Delta \mathbf{A}} \preceq \Delta \mathbf{A} \preceq \overline{\Delta \mathbf{A}}, \tag{5.25a}$$

$$\underline{\Delta \mathbf{B}} \preceq \Delta \mathbf{B} \preceq \overline{\Delta \mathbf{B}} \tag{5.25b}$$

or equivalently

$$\mathbf{A} + \underline{\Delta \mathbf{A}} \preceq \mathbf{A} + \Delta \mathbf{A} \preceq \mathbf{A} + \overline{\Delta \mathbf{A}}, \tag{5.26a}$$

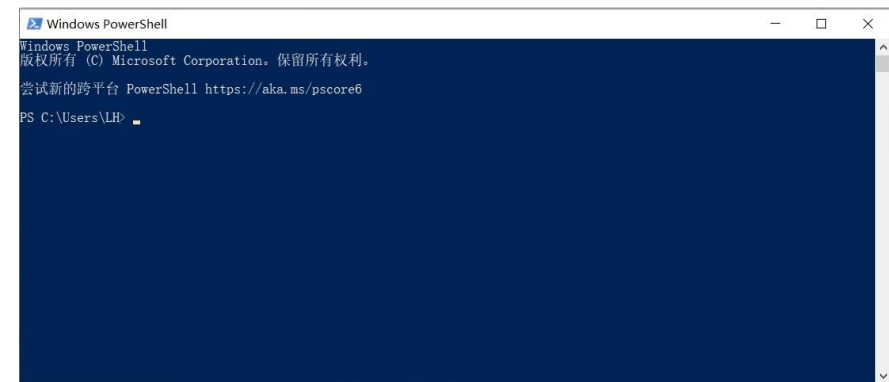
$$\mathbf{B} + \underline{\Delta \mathbf{B}} \preceq \mathbf{B} + \Delta \mathbf{B} \preceq \mathbf{B} + \overline{\Delta \mathbf{B}}. \tag{5.26b}$$

Principles

- Determine Users' Skill Levels
 - When multiple user classes must be accommodated, the basic strategy is to permit a **multi-layer (or level-structured) approach** to learning
 - Novice users can be taught a minimal subset of objects and actions to get started with
 - Middle-level users can choose to progress to ever-greater levels of task and UI concepts
 - For users with strong knowledge of task and UI concepts, rapid progress is possible
 - The multi-layer approach helps users with different skill levels and promotes universal usability



Windows GUI for novice users



Windows PowerShell for advanced users

Principles

- Determine Users' Skill Levels
 - When multiple user classes must be accommodated, the basic strategy is to permit a **multi-layer (or level-structured) approach** to learning
 - Novice users can be taught a minimal subset of objects and actions to get started with
 - Middle-level users can choose to progress to ever-greater levels of task and UI concepts
 - For users with strong knowledge of task and UI concepts, rapid progress is possible
 - The multi-layer approach helps users with different skill levels and promotes universal usability
 - Another option for accommodating different user classes is to permit users to personalize the menu contents
 - A third option is to permit users to control the density and information feedback that the UI provides

Principles

- Identify the Tasks
 - After carefully drawing the user profile, UIDers identify the tasks to be carried out, successful strategies of which usually involve long hours of observing and interviewing users
 - Identifying the tasks helps UIDers to understand **task frequencies/sequences** and make the tough decisions about what tasks to support
 - Bad E.g. Some UIDers prefer to include all possible actions in the hope that some users will find them helpful, but this causes clutter --- Reflect which usability goal is violated?
 - High-level task actions can be decomposed into multiple middle-level task actions, which can be further refined into **atomic actions** that users execute with a single menu selection or other action
 - Choosing the most appropriate set of atomic actions is indeed a **difficult** task

Sensitive observation

- Openness, curiosity, ...



Dialectic (positive<->negative) thinking

- Inverse thinking, perspective taking, essence-grasping thinking, ...

Principles

- Identify the Tasks
 - **Relative task frequencies** are important in shaping a menu tree and are one of the bases for making architectural design decisions
 - Reflect what other bases are there for making architectural design decisions?
 - Frequent tasks should be near the top and therefore quick to carry out, while rare tasks are deeper down
 - (In a text editor) Frequent actions might be performed by pressing special keys
 - Less frequent actions might be performed by pressing a single letter plus the Ctrl key or by a single selection from a pull-down menu
 - Infrequent actions or complex actions might require going through a sequence of menu selections or form fill-ins (E.g. change the margins or revise default printers)
 - Creating a users-tasks matrix (with simple check marks or more precise statistics/analysis of frequencies) can help UIDers sort out these issues

Principles

How to take relative task frequencies into account in UID from the engineering perspective?

- Identify the Tasks
 - **Relative task frequencies** are important in shaping a menu tree and are one of the bases for making architectural design decisions
 - Such user-needs assessment clarifies what tasks are essential for the UID and which ones could be left out to preserve system simplicity and learning ease

	TASK				
Job Title	Query by Patient	Update Data	Query across Patients	Add Relations	Evaluate System
Nurse	**	**			
Physician	**	*			
Supervisor	*	*	**		
Appointment personnel	*****				
Medical-record maintainer	**	**	*	*	
Clinical researcher			***		*
Database programmer		*	**	**	*

A users-tasks table with simple check marks, yet more precise statistics/analysis of frequencies can be incorporated

Principles

- Choose an Interaction Style
 - When the task analysis is complete and the task objects/actions are identified, the UIDer can choose from five primary interaction styles
 - Direct manipulation
 - A knowledge of CV/CG do can help a lot to UID
 - Navigation and menu selection
 - Form fill-in
 - Natural language
 - Command language
 - Command languages provide users a strong feeling of being in control and enable users to express complex possibilities rapidly without bothering to read distracting prompts

Advantages	Disadvantages
Direct manipulation • Visually presents task concepts • Allows easy learning • Allows easy retention • Allows errors to be avoided • Encourages exploration • Affords high subjective satisfaction	• May be hard to program • Accessibility requires special attention
Navigation and menu selection • Shortens learning • Reduces keystrokes • Structures decision making • Permits use of dialog-management tools • Allows easy support of error handling	• Presents danger of many menus • May slow frequent users • Consumes screen space • Requires rapid display rate
Form fill-in • Simplifies data entry • Enables convenient assistance • Permits use of form-management tools	• Consumes screen space
Command language • Powerful • Allows easy scripting and history keeping	• Requires learning and retention • Error-prone
Natural language • Relieves burden of learning syntax	• Requires clarification dialog • May not show context • May require more keystrokes • Unpredictable

Principles

- Eight Golden Rules of UID
 - Strive for consistency
 - Consistent sequences of actions should be required in similar situations
 - Identical terminology should be used in prompts, menus, and help screens
 - Consistent colour, layout, capitalization, fonts, and so on should be employed throughout
 - Exceptions such as confirmation of the delete command should be comprehensible
 - Seek universal usability
 - Recognize the needs of diverse users and design for plasticity, facilitating transformation of content
 - Novice to expert differences, age ranges, disabilities, international variations, technological diversity, and so on, each enrich the spectrum of requirements that can guide UID

“There is no medicine against death, and against error no rule has been found”

Sigmund Freud (inscription he wrote on his portrait)

人非圣贤，孰能无过

Principles

- Eight Golden Rules of UID
 - Offer informative feedback
 - For every user action, there should be an UI feedback
 - Responses for frequent and minor actions can be modest
 - Responses for infrequent and major actions should be more substantial
 - Take advantage of visualization that brings convenience to showing changes explicitly
 - Design dialogs to yield closure
 - Sequences of actions should be organized into groups with a beginning, middle, and end
 - Informative feedback at the completion of a group of actions gives users satisfaction of accomplishment, a sense of relief, and an indicator to prepare for the next action group
 - Prevent errors
 - Make as sure as possible that users cannot make serious errors during UI usage
 - If users make an error, the UI should offer simple, constructive, and specific instructions for recovery
 - Erroneous actions should leave the UI state unchanged and recoverable

Principles

- Eight Golden Rules of UID
 - Permit easy reversal of actions
 - Enable actions as much as possible to be reversible, which relieves anxiety
 - Reversibility units may be a single action, a data-entry, or a complete group of actions
 - Keep users in control ($\neq$ in the control)
 - Experienced users strongly desire the sense that they are in charge of the UI and that the UI responds to their actions
 - Reduce short-term memory load
 - Humans' limited capacity for information processing in short-term memory requires that UIDers avoid UIs in which users must remember information for inter-display usage
 - These underlying principles, which have their limitations but provide a good starting point for UID, must be interpreted, refined, and extended for each environment

持经达变

Principles

- Ensure **Human Control** while Increasing Automation

Humans Generally Better

- Sense-making from hearing, sight, touch, etc.
- Detect familiar signals in noisy background
- Draw on experience and adapt to situations
- Select alternatives if original approach fails
- Act in unanticipated situations
- Apply principles to solve varied problems
- Make subjective value-based judgments
- Develop new solutions
- Use information from external environment
- Request help from other humans

Machines Generally Better

- Sense stimuli outside human's range
- Rapid consistent response for expected events
- Retrieve detailed information accurately
- Process data with anticipated patterns
- Perform repetitive actions reliably
- Perform several activities simultaneously
- Maintain performance over time